

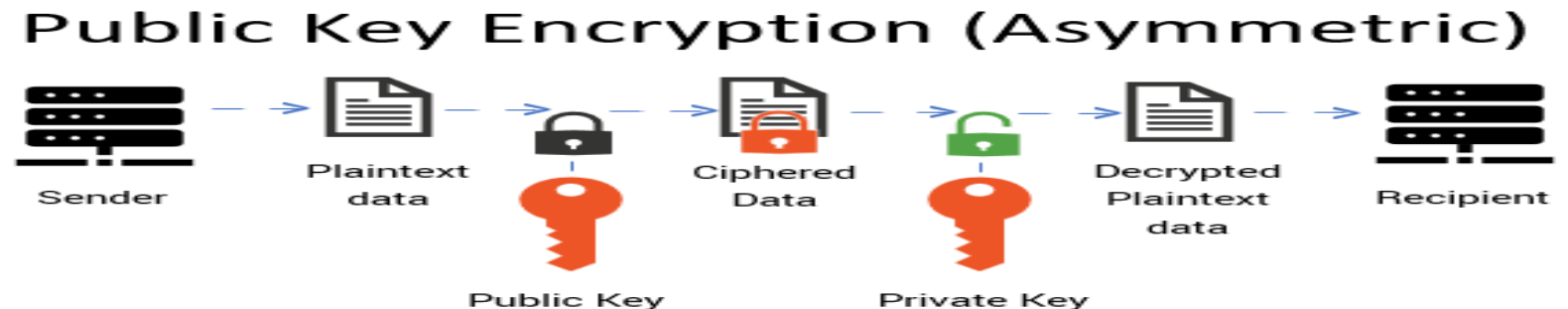
Cyber Security & Laws

BCAC 0032

Module-II

Public key cryptography

- In 1976, Whitfield Diffie and Martin Hellman introduced the concept of Public Key cryptography (asymmetric cryptography).
- In their system, each person is the owner of a mathematically related pair of keys: a Public Key, intended to be available to anyone who wants it; and a Private Key, which is kept secret and only known by the owner.
- Because messages are encrypted with a Public Key and can only be decrypted by the related Private Key.
- In contrast to symmetric key systems, public key systems require each user **A** to have two keys: a public key, $K_{pub}(A)$ that is published and a private key $K_{pri}(A)$ that is kept secret. A message encrypted (with a ruleset **E**) using one key can be decrypted (using ruleset **D**) using only the other key.



Public key Encryption Algorithms

- **RSA Cryptosystem**
- **Diffie-Hellman Key Exchange**
- **Elliptic Curve Cryptography**

RSA Algorithms

- RSA is the best known, and by far the most widely used general public key encryption algorithm, and was first published by Rivest, Shamir & Adleman of MIT in 1978.
- RSA involves a public-key and a private-key where the public key is known to all and is used to encrypt data or message. The data or message which has been encrypted using a public key can only be decrypted by using its corresponding private-key. Each user generates a key pair i.e. public and private key using the following steps:
 - Selects two large primes at random - p, q
 - compute the modulus $n=p.q$
 - calculate $\phi(n)$, where $\phi(n)=(p-1)(q-1)$
 - selecting at random the encryption key e , where $1 < e < \phi(n)$, and $\gcd(e, \phi(n))=1$
 - solve following equation to find decryption key d : $e.d=1 \pmod{\phi(n)}$ and $0 \leq d \leq n$
 - publish their public encryption key: $KU=\{e,n\}$
 - keep secret private decryption key: $KR=\{d,n\}$

- Both the sender and receiver must know the values of n and e , and only the receiver knows the value of d . Encryption and Decryption are done using the following equations.
- To encrypt a message M the sender:
 - obtains public key of recipient $KU=\{e,n\}$
 - computes: $C=M^e \bmod n$, where $0 \leq M$
- To decrypt the ciphertext C the owner:
 - uses their private key $KR=\{d,n\}$
 - computes: $M=C^d \bmod n = (M^e)^d \bmod n = M^{ed} \bmod n$

Example Calculation

1. **Choose Primes:** $p = 7, q = 19$.
2. **Calculate n :** $n = pq = 7 \times 19 = 133$.
3. **Calculate $\phi(n)$:** $\phi(n) = (p - 1)(q - 1) = 6 \times 18 = 108$.
4. **Choose e :** Choose e such that $1 < e < 108$ and $\gcd(e, 108) = 1$. Let $e = 5$.
5. **Calculate d :** d such that $5d \equiv 1 \pmod{108}$. A possible value for d is 65 (since $5 \times 65 = 325$, and $325 \pmod{108} = 1$).
6. **Encryption:** $C = M^e \pmod{n}$.
7. **Decryption:** $M = C^d \pmod{n}$. 

Deffie-Hellman Key Exchange

- The Diffie-Hellman (DH) algorithm is a foundational cryptographic protocol (1976) that allows two parties to securely establish a shared secret key over an insecure communication channel, without transmitting the key itself.
- It enables secure, symmetric encrypted communication between parties that have no prior knowledge of each other.

Diffie-Hellman Key Exchange Algo

1. **Public Parameters:** Alice and Bob publicly agree on a large prime number p and a primitive root modulo p , g .
2. **Private Keys:** Alice chooses a secret private key a . Bob chooses a secret private key b .
3. **Public Key Calculation:**
 - Alice computes public key $A = g^a \pmod{p}$.
 - Bob computes public key $B = g^b \pmod{p}$.
4. **Exchange:** Alice sends A to Bob, and Bob sends B to Alice.
5. **Shared Secret Generation:**
 - Alice computes shared secret $K = B^a \pmod{p} = (g^b)^a \pmod{p}$.
 - Bob computes shared secret $K = A^b \pmod{p} = (g^a)^b \pmod{p}$.

Both parties now possess the same secret $K = g^{ab} \pmod{p}$

Numerical

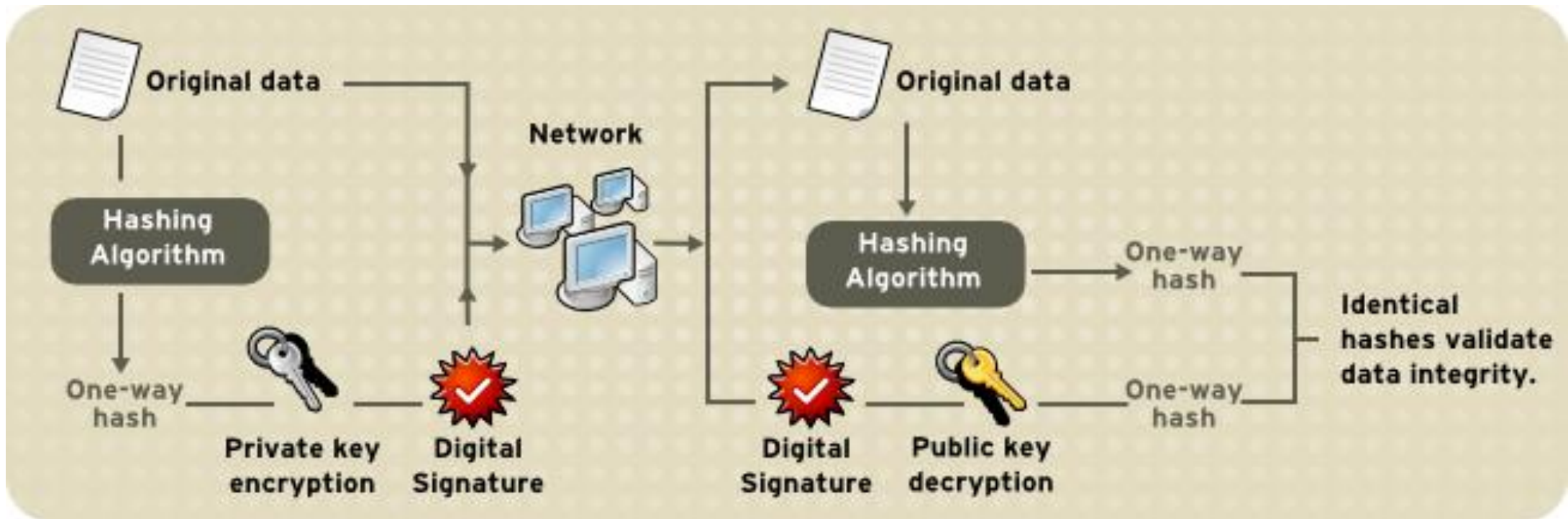
- **Public Values:** $p = 23, g = 5$
- **Private Values:** Alice $a = 4$, Bob $b = 3$
- **Public Exchange:**
 - Alice sends $A = 5^4 \pmod{23} = 625 \pmod{23} = 4$
 - Bob sends $B = 5^3 \pmod{23} = 125 \pmod{23} = 10$
- **Secret Calculation:**
 - Alice computes $K = B^a \pmod{p} = 10^4 \pmod{23} = 18$
 - Bob computes $K = A^b \pmod{p} = 4^3 \pmod{23} = 18$
- **Result:** Shared Secret = 18. 

Applications of Public-key Cryptosystems

- **Authentication-** Authentication is the process of verifying one's identity.
- **Key Exchange-** The requirement is to share a key, which can later be used to encrypt and decrypt messages by a secret key system like DES.
- **Confidentiality-** Confidentiality means protecting data from eavesdroppers.
- **Non-repudiation-** This requirement comes from the issue of denying previous actions.
- **Digital Signatures-** The concept is much similar to the real world process of signing a document to make it legal and to specify the person responsible for the document.
- **Data Integrity-** Digital signatures can also be used to validate the integrity of signed data.

Digital Signature

A digital signature is a cryptographic mechanism using [asymmetric cryptography](#) to verify the authenticity, integrity, and non-repudiation of digital messages or documents.



Working of Digital Signature

- **How it Works:** The sender hashes the document and encrypts the hash with their private key, creating a unique signature. The recipient decrypts this with the sender's public key and verifies the hash matches.
- **Key Components:**
 - **Hash Function:** Produces a unique digital fingerprint of the data.
 - **Private key:** Used by the sender to sign the hash.
 - **Public key:** Used by anyone to verify the signature.

Applications of DS

1. Secure Email Communication
2. E-Commerce Transactions
3. Electronic Documents and Contracts
4. E-Governance Services
E-tendering and e-procurement
5. Software Distribution
6. Banking and Financial Transactions
7. Secure Data Transmission
8. Electronic Voting Systems
9. Healthcare Systems

Digital watermarking

- **Digital watermarking** is a technique used to embed hidden information (called a **watermark**) into digital media such as :
 - images,
 - audio,
 - video, or
 - documents.
- The embedded information is used to **verify ownership, authenticate content, and prevent unauthorized copying or distribution.**
- The watermark is usually invisible to users but can be detected or extracted using specific algorithms.

Features/characteristics of Digital Watermarking

- 1. Imperceptibility** – The watermark should not affect the quality of the original content.
- 2. Robustness** – The watermark should remain detectable even after compression, editing, or noise.
- 3. Security** – Unauthorized users should not be able to remove or modify the watermark.
- 4. Capacity** – Ability to embed sufficient information without degrading the content.

5. Reliability

- The watermark detection system should be **accurate and reliable**, meaning it should correctly identify the presence or absence of the watermark.

6. Computational Efficiency

- The watermark embedding and extraction processes should require **reasonable computational resources** so that they can be applied in real-time applications.

7. Unambiguity

- The watermark should **clearly identify the owner or source** of the digital content without confusion or false detection.

Types of Digital Watermarking

1. Visible Watermarking

- The watermark is **visible to users**.
- Commonly used in images and videos to indicate ownership.

Examples:

- TV channel logos on broadcasts
- Watermarks on stock photos

2. Invisible Watermarking

- The watermark is **hidden within the media** and cannot be seen by the human eye.
- It is detected using special software.
- Used for copyright protection and content authentication.

Examples:

- Broadcast monitoring
- Transaction tracking

3. Fragile Watermarking

- Designed to detect **tampering or modification** in digital content.
- If the content is changed, the watermark becomes invalid.

Examples:

- Used in **authentication systems**.

4. Robust Watermarking

- Designed to survive **compression, resizing, filtering, and noise**.
- They are difficult to remove or disable without degrading quality.

Examples:

- Mainly used for **copyright protection**.

Working of Digital Watermarking

The process of watermarking involves three main steps.

- **Watermarking Generation-** The watermark is generated based on application requirements like robustness, security, etc.
- **Watermarking Embedding-** Watermark Embedded into the original digital media using an embedding algorithm.
- **Watermarking Extraction/Detection-** Extraction algorithm is designed specially to counter transformations like compression, noise, filtering, etc.